Lab 10

Aim	Implementation of statistical functions to understand the data in R.

Theory	The theory provided covers fundamental statistical methods and hypothesis testing that can be conducted in R. Below is a detailed breakdown of each of the key concepts mentioned:
	
	1. Basic Descriptive Statistics
	
	Descriptive statistics summarize and describe the main features of a dataset, often in terms of its central tendency, dispersion, and distribution.
	
	• **Mean (Average)**:
	  o The mean is the average of all values in a dataset. It's calculated by summing all the values and dividing by the number of values.
	• **Median (Middle Value)**:
	  o The median represents the middle value of a dataset when arranged in order. If there is an even number of values, it is the average of the two middle values.
	• **Mode (Most Frequent Value)**:
	  o The mode is the value that appears most frequently in the dataset. R doesn't have a built-in mode function, but it can be created using the `unique()` and `tabulate()` functions.
	• **Variance (Spread of Data)**:
	  o Variance measures how far individual data points are from the mean. A larger variance indicates greater variability in the dataset.
	• **Standard Deviation (Spread of Data)**:
	  o Standard deviation is the square root of variance. It's often more interpretable than variance because it is in the same units as the data.
	• **Summary Statistics**:
	  o The `summary()` function in R provides key descriptive statistics, including the minimum, first quartile, median, mean, third quartile, and maximum values.
	
	2. Measuring Data Spread
	
	• **Range (Max - Min)**:
	  o The range is simply the difference between the maximum and minimum values in the dataset. It gives a rough idea of how spread out the data is.
	• **Interquartile Range (IQR)**:
	  o The IQR is the difference between the third quartile (Q3) and first quartile (Q1), which captures the spread of the middle 50% of the data. It is useful for identifying outliers.

3. Skewness & Kurtosis

- **Skewness (Symmetry of Data)**:
  - Skewness measures the asymmetry of the data. If the data is skewed to the right (positively skewed), the tail is on the right side, and if it is skewed to the left (negatively skewed), the tail is on the left side.
- **Kurtosis (Shape of Data)**:
  - Kurtosis measures the "tailedness" of the distribution. High kurtosis indicates heavy tails or outliers, while low kurtosis indicates light tails.

4. Correlation & Covariance

- **Correlation (Strength of Relationship)**:
  - Correlation measures the strength and direction of a linear relationship between two variables. A correlation value close to 1 or -1 indicates a strong linear relationship, while a value near 0 suggests no linear relationship.
- **Covariance (Direction of Relationship)**:
  - Covariance is similar to correlation, but it is not standardized. It measures the direction of the relationship between two variables—whether they move together (positive covariance) or in opposite directions (negative covariance).

5. Normality Tests

- **Shapiro-Wilk Test (Check if Data is Normally Distributed)**:
  - The Shapiro-Wilk test is a formal statistical test to determine whether a dataset follows a normal distribution. If the p-value is small (typically less than 0.05), we reject the hypothesis of normality.
- **Q-Q Plot (Visual Inspection of Normality)**:
  - A Q-Q plot compares the quantiles of the dataset to the quantiles of a normal distribution. If the points lie roughly along a straight line, the data is approximately normal.

6. Hypothesis Testing

Hypothesis testing is used to evaluate assumptions about a population based on sample data. It involves formulating a null hypothesis and an alternative hypothesis, followed by testing the hypothesis using statistical methods.

- **t-test (Compare Two Means)**:
  - A t-test compares the means of two groups to determine if there is a significant difference between them. The result provides a p-value, and if it’s smaller than a significance level (commonly 0.05), the null hypothesis is rejected.
- **Chi-Square Test (Check Independence of Categorical Data)**:
  - The chi-square test is used to determine if there is an association between two categorical variables. It compares observed frequencies with expected frequencies and calculates a chi-square statistic.

Conclusion:

R provides a wide array of functions for performing statistical analysis. Whether you're calculating basic statistics, measuring the spread of data, examining data distribution, or conducting hypothesis testing, R's functions and packages allow for efficient and powerful data analysis.

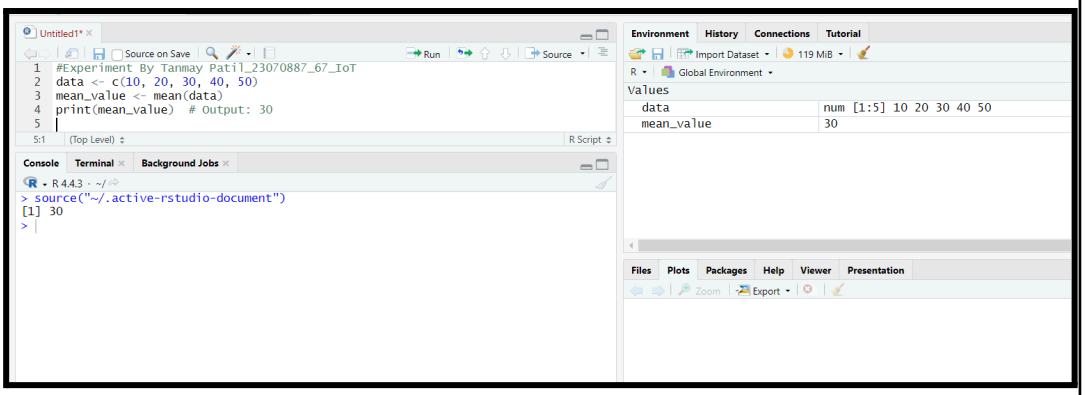
Program

```
i) data <- c(10, 20, 30, 40, 50)
mean_value <- mean(data)
print(mean_value)  # Output: 30


ii) median_value <- median(data)
print(median_value)  # Output: 30


iii) mode_function <- function(x) {
  uniq_vals <- unique(x)
  uniq_vals[which.max(tabulate(match(x, uniq_vals)))]
}
data2 <- c(1, 2, 2, 3, 3, 3, 4, 4)
mode_value <- mode_function(data2)
print(mode_value)  # Output: 3


iv) var_value <- var(data)
print(var_value)  # Output: 250


v) sd_value <- sd(data)
print(sd_value)  # Output: 15.81


vi) range_value <- range(data)
print(range_value)  # Output: 10 50
range_diff <- diff(range_value)
print(range_diff)  # Output: 40


vii) iqr_value <- IQR(data)
print(iqr_value)  # Output: 20


viii) library(moments)
skew_value <- skewness(data)
print(skew_value)


ix) kurt_value <- kurtosis(data)
print(kurt_value)
```

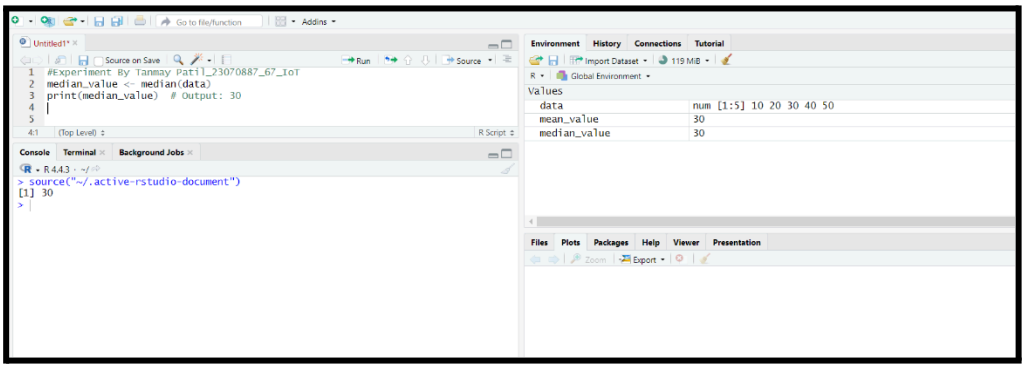
```
x) x <- c(1, 2, 3, 4, 5)
y <- c(2, 4, 6, 8, 10)
correlation <- cor(x, y)
print(correlation)  # Output: 1


xi) covariance <- cov(x, y)
print(covariance)  # Output: 5


xii) shapiro.test(data)


xiii) qqnorm(data)
qqline(data, col = "red")


xiv) group1 <- c(12, 15, 14, 10, 18)
group2 <- c(22, 25, 21, 24, 27)
t_test <- t.test(group1, group2)
print(t_test)


xv) data_matrix <- matrix(c(10, 20, 30, 40), nrow = 2)
chi_test <- chisq.test(data_matrix)
print(chi_test)
```

Output

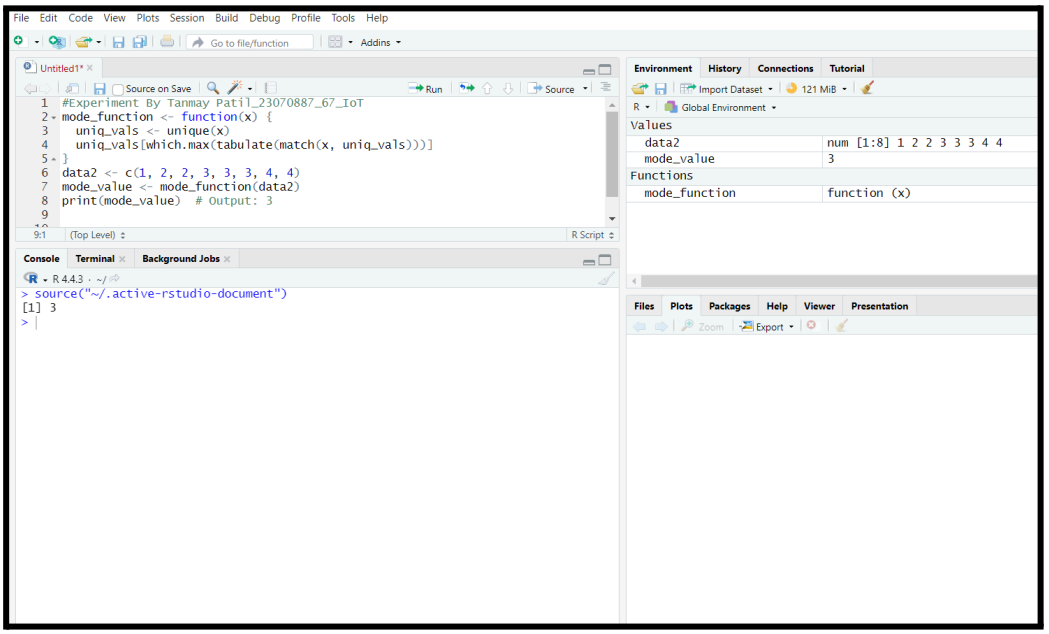
i)



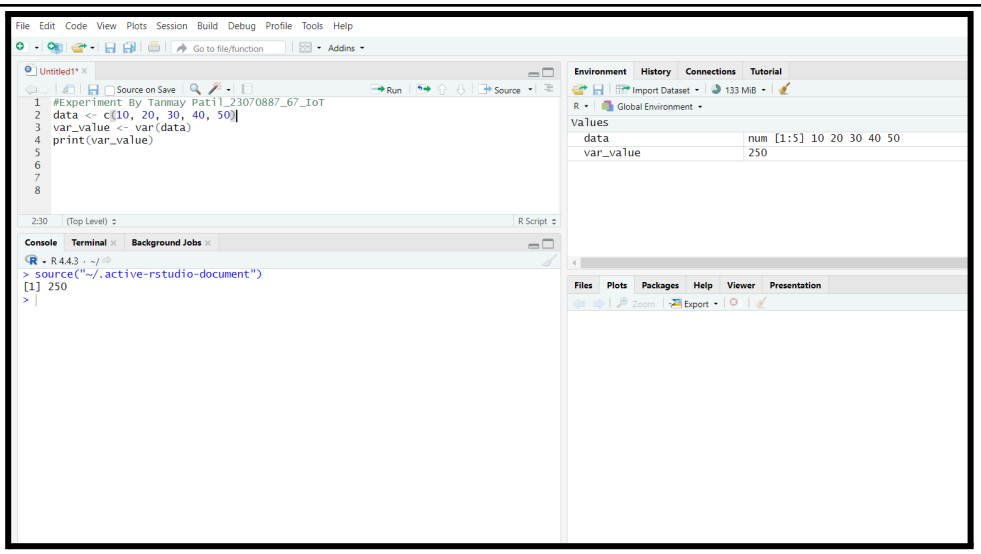
ii)


#Experiment By Tanmay Patil_23070887_67_IoT
median_value <- median(data)
print(median_value) # Output: 30
> source("~/.active-rstudio-document")
[1] 30
Values
data
num [1:5] 10 20 30 40 50
mean_value
30
median_value
30

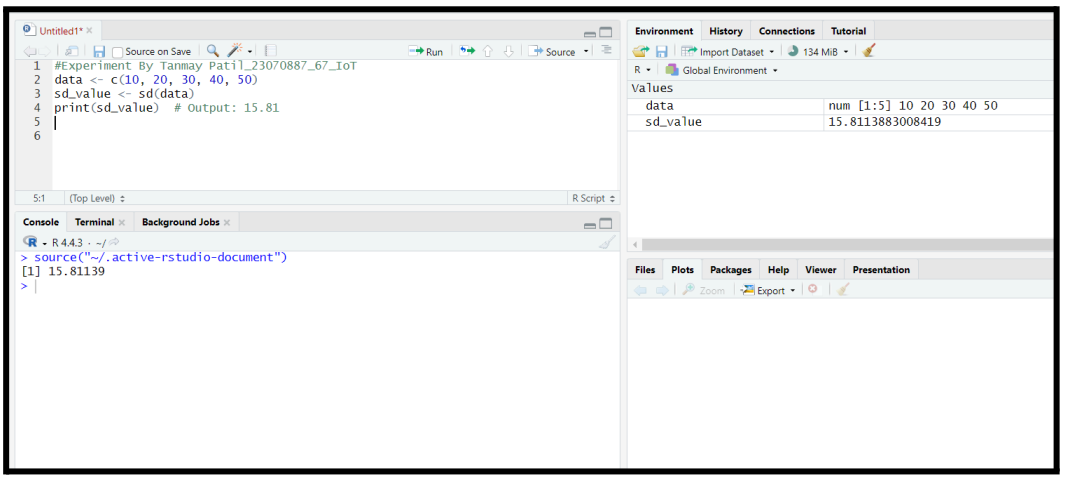
iii)


#Experiment By Tanmay Patil_23070887_67_IoT
mode_function <- function(x) {
uniq_vals <- unique(x)
uniq_vals[which.max(tabulate(match(x, uniq_vals)))]
}
data2 <- c(1, 2, 2, 3, 3, 3, 4, 4)
mode_value <- mode_function(data2)
print(mode_value) # Output: 3
> source("~/.active-rstudio-document")
[1] 3
Values
data2
num [1:8] 1 2 2 3 3 3 4 4
mode_value
3
Functions
mode_function
function (x)

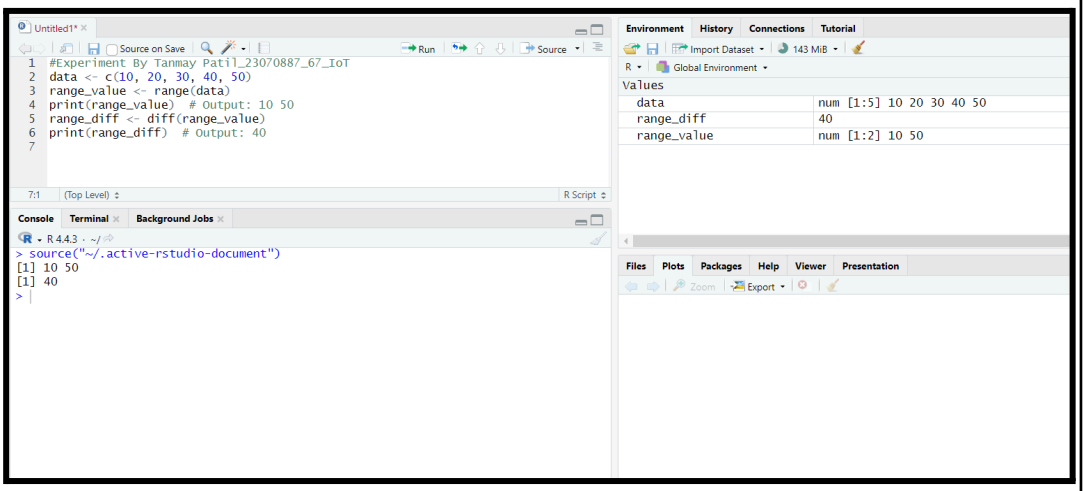
iv)


#Experiment By Tanmay Patil_23070887_67_IoT
data <- c(10, 20, 30, 40, 50)
var_value <- var(data)
print(var_value)
> source("~/.active-rstudio-document")
[1] 250
Values
data num [1:5] 10 20 30 40 50
var_value 250

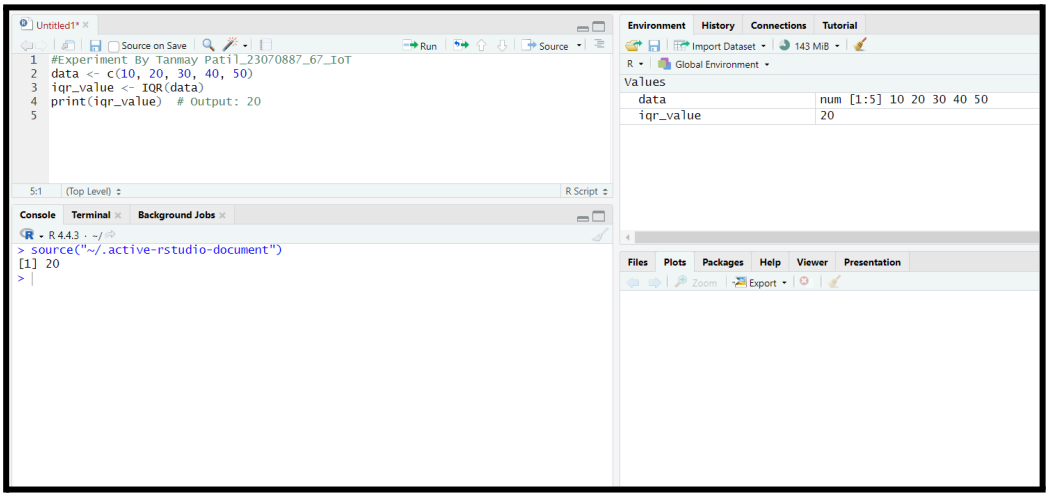
v)


#Experiment By Tanmay Patil_23070887_67_IoT
data <- c(10, 20, 30, 40, 50)
sd_value <- sd(data)
print(sd_value) # Output: 15.81
> source("~/.active-rstudio-document")
[1] 15.81139
Values
data num [1:5] 10 20 30 40 50
sd_value 15.811883008419

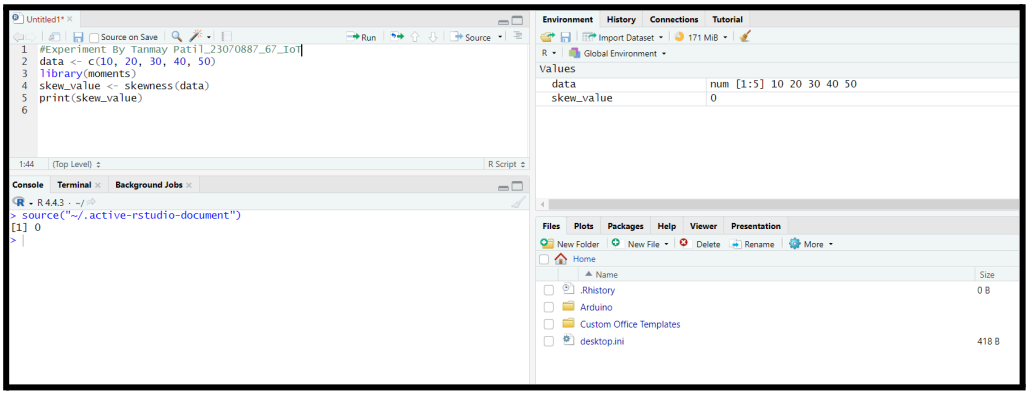
vi)


#Experiment By Tanmay Patil_23070887_67_IoT
data <- c(10, 20, 30, 40, 50)
range_value <- range(data)
print(range_value) # Output: 10 50
range_diff <- diff(range_value)
print(range_diff) # Output: 40
> source("~/.active-rstudio-document")
[1] 10 50
[1] 40
Values
data num [1:5] 10 20 30 40 50
range_diff 40
range_value num [1:2] 10 50

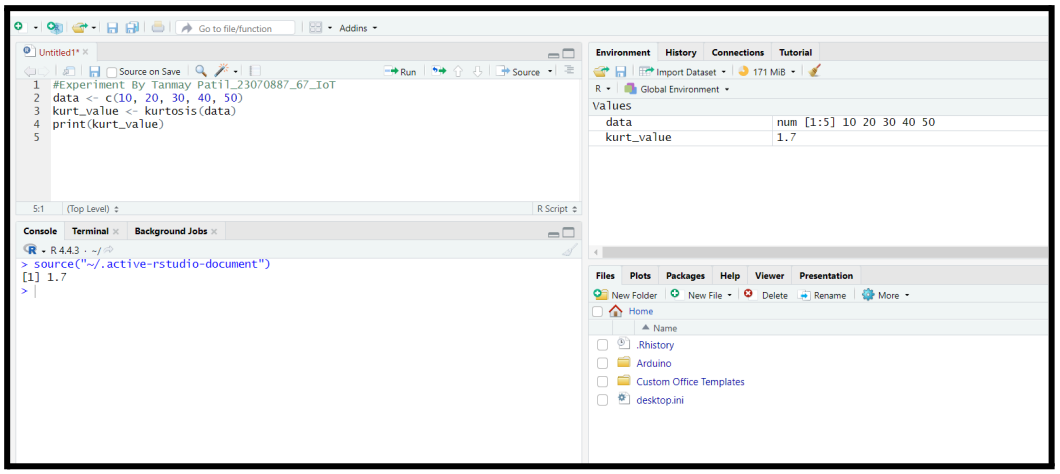
vii)


#Experiment By Tanmay Patil_23070887_67_IoT
data <- c(10, 20, 30, 40, 50)
iqr_value <- IQR(data)
print(iqr_value) # Output: 20
> source("~/.active-rstudio-document")
[1] 20
Values
data num [1:5] 10 20 30 40 50
iqr_value 20

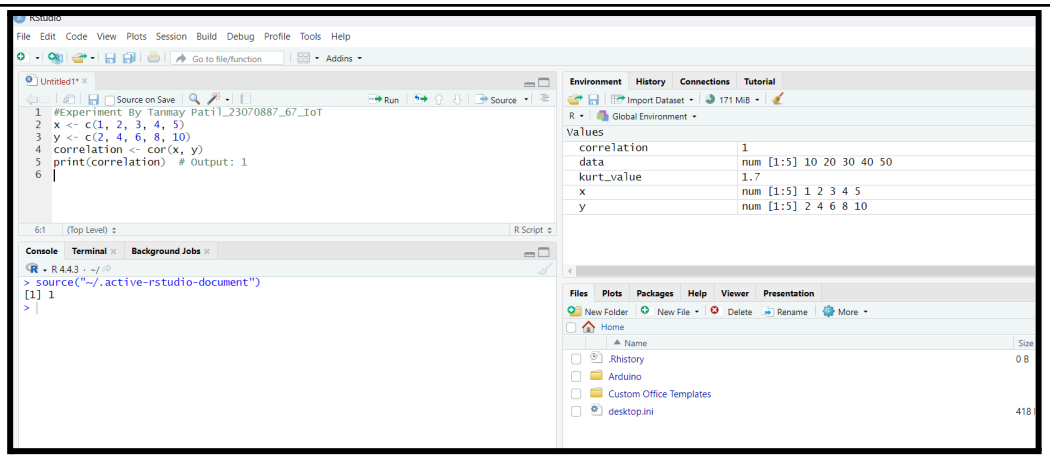
viii)


#Experiment By Tanmay Patil_23070887_67_IoT
data <- c(10, 20, 30, 40, 50)
library(moments)
skew_value <- skewness(data)
print(skew_value)
> source("~/.active-rstudio-document")
[1] 0
Values
data num [1:5] 10 20 30 40 50
skew_value 0

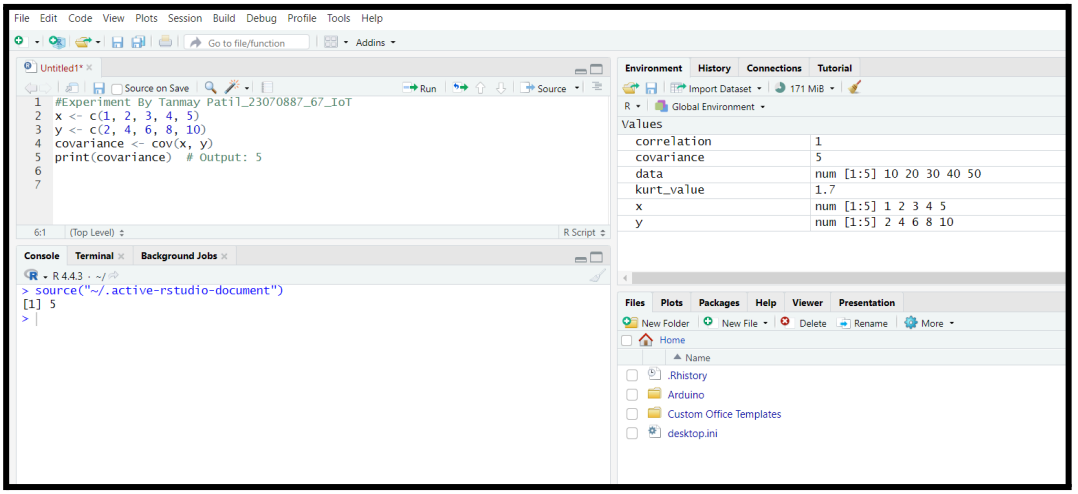
ix)


#Experiment By Tanmay Patil_23070887_67_IoT
data <- c(10, 20, 30, 40, 50)
kurt_value <- kurtosis(data)
print(kurt_value)
> source("~/.active-rstudio-document")
[1] 1.7
Values
data num [1:5] 10 20 30 40 50
kurt_value 1.7

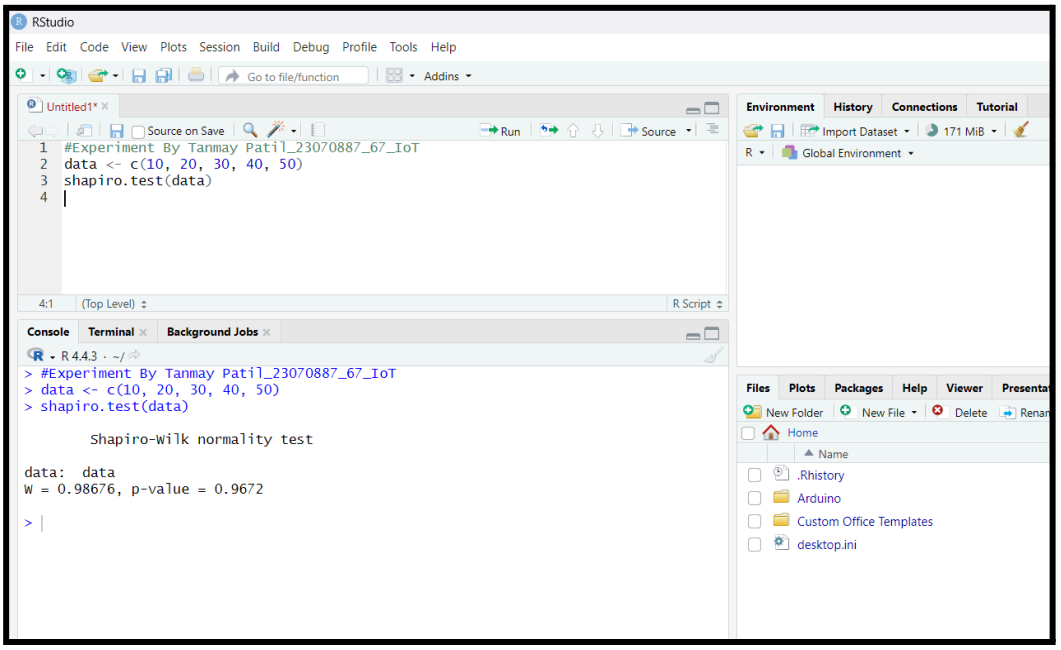
x)


#Experiment By Tanmay Patil_23070887_67_IoT
x <- c(1, 2, 3, 4, 5)
y <- c(2, 4, 6, 8, 10)
correlation <- cor(x, y)
print(correlation) # Output: 1
> source("~/.active-rstudio-document")
[1] 1
correlation 1
data num [1:5] 10 20 30 40 50
kurt_value 1.7
x num [1:5] 1 2 3 4 5
y num [1:5] 2 4 6 8 10

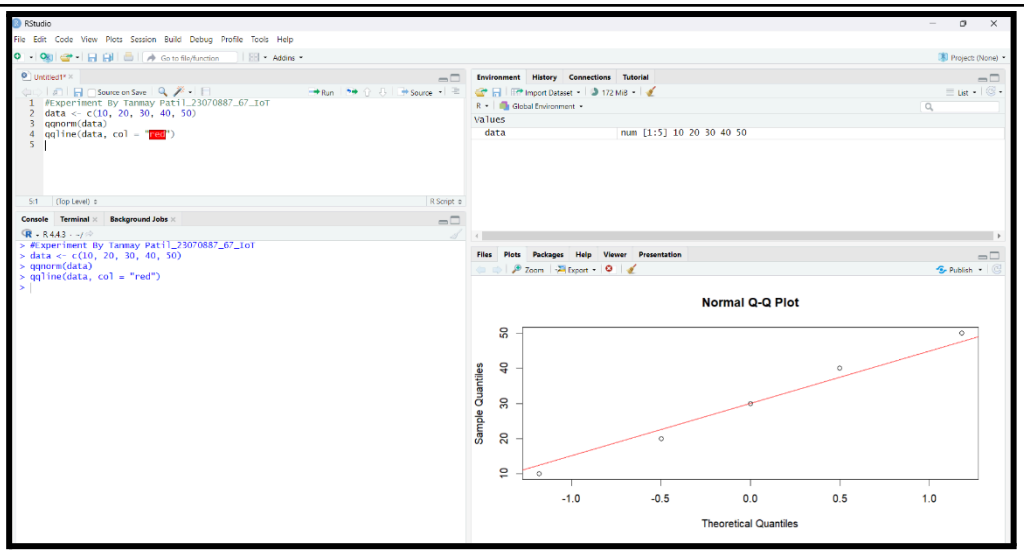
xi)


#Experiment By Tanmay Patil_23070887_67_IoT
x <- c(1, 2, 3, 4, 5)
y <- c(2, 4, 6, 8, 10)
covariance <- cov(x, y)
print(covariance) # Output: 5
> source("~/.active-rstudio-document")
[1] 5
correlation 1
covariance 5
data num [1:5] 10 20 30 40 50
kurt_value 1.7
x num [1:5] 1 2 3 4 5
y num [1:5] 2 4 6 8 10

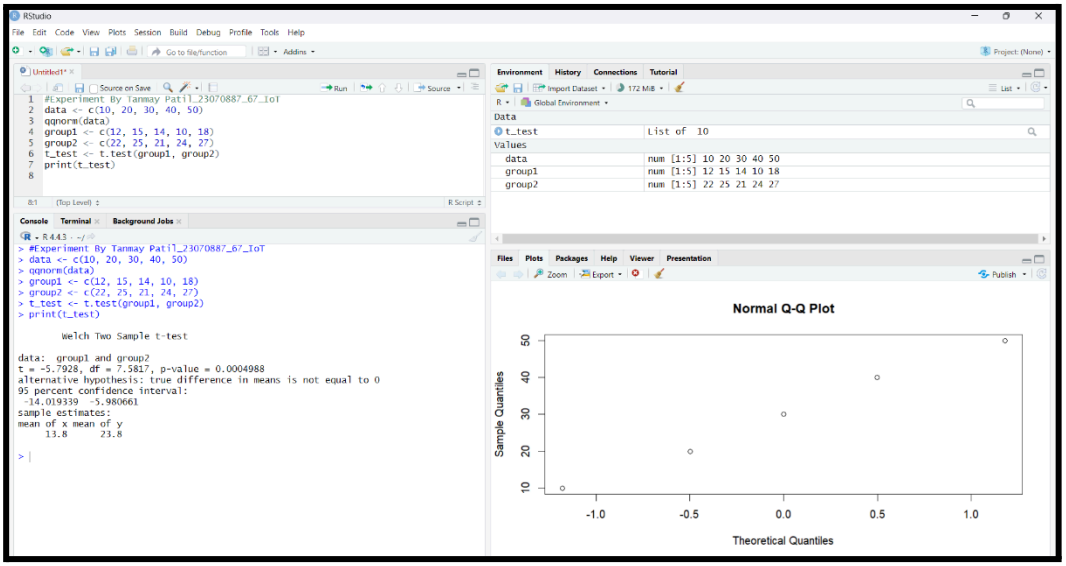
xii)


#Experiment By Tanmay Patil_23070887_67_IoT
data <- c(10, 20, 30, 40, 50)
shapiro.test(data)
> #Experiment By Tanmay Patil_23070887_67_IoT
> data <- c(10, 20, 30, 40, 50)
> shapiro.test(data)
Shapiro-Wilk normality test
data: data
W = 0.98676, p-value = 0.9672

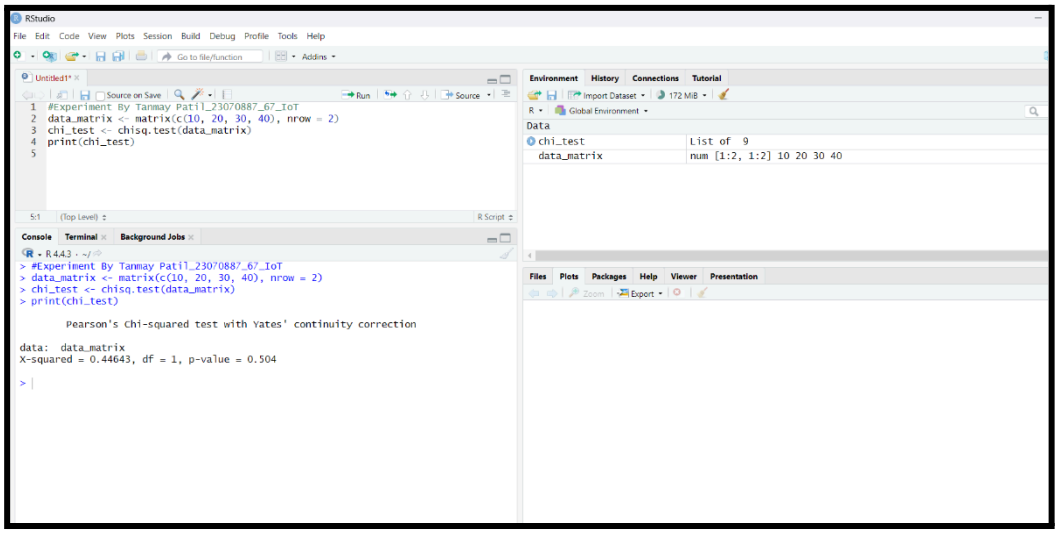
xiii)



xiv)



xv)



Conclusion	Implementation of statistical functions to understand the data in R executed successfully.

Name of Faculty	Akhil Jajulwar Sir.